Almost Gorenstein numerical semigroup rings which are almost complete intersection

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Aim of this talk

The **2** Keywords of this talk are
Almost Gorenstein (Almost Symmetric) and **Almost Complete Intersection**.
Let $(S, \mathfrak{n})$ be a regular local ring and $R = S/I$, with height $I = h$. R is said to be an **Almost Complete Intersection (ACI)** if I is minimally generated by $h + 1$ elements. The following theorem of Kunz is very famous.

Theorem. If R is an ACI, then R is **not** Gorenstein.
Let $H = \langle n_1, n_2, \ldots, n_e \rangle$ be a numerical semigroup. The following Theorem is also due to Kunz.
Theorem The semigroup ring $k[H]$ is Gorenstein if and only if H is symmetric.
Now we ask, how about if we change “Gorenstein” to “almost Gorenstein” or, equivalently, ”symmetric” to “almost symmetric”.
This is the aim of this talk and the answer is
ACI are not almost Gorenstein, **except for a series of examples, described by a special shape of RF** (Row Factorization) **matrices.**
In the following, let $H = \langle n_1, n_2, \ldots, n_e \rangle$ be a numerical semigroup minimally generated by $e \geq 3$ elements. We denote $S = k[X_1, \ldots, X_e]$ be a polynomial ring over a field k so that $k[H] = S/I_H$.

1 dimensional almost Gorenstein ring was introduced by Barucci and Fröberg, in [BF] in 1997. For simplicity, we define it only for $k[H]$.

Definition. $k[H]$ is almost Gorenstein if and only if for every $x \in \mathbb{Z} \setminus H$, $\mathrm{F}(H) - x + H_+ \subset H$. If H satisfies the latter condition, H is said to be **almost symmetric**.
Almost symmetric property is described by **pseudo-Frobenius numbers**.
Definition. $\mathrm{PF(H)} = \{x \in \mathbb{Z} \setminus H \mid x + H_+ \subset H\}$.
Theorem (Nari, [N13]) Let $\mathrm{PF}(H) = \{f_1 < f_2 < \ldots < f_t = \mathrm{F}(H)\}$. Then H is almost symmetric if and only if $f_i + f_{t-i} = \mathrm{F}(H)$ holds for $t = 1, 2, \ldots, t-1$.
The property "almost Gorenstein" does not hold if H is given by "gluing" of H_1 and H_2. ($H = \langle aH_1, bH_2 \rangle$ with GCD(a,b)=1, $a \in H_2, b \in H_1, a, b > 1$ including $H_i = \mathbb{N}$) This fact plays an important role in our results.
Theorem (Nari, [N13]) If H is the gluing of H_1 and H_2 and if H is not symmetric, then H is *not* almost symmetric.

First Main Theorem

Main Thorem 1. Assume that H is almost symmetric, not symmetric, and that $k[H]$ is an almost complete intersection. Then

1. H is pseudo-symmetric
2. $e = 2e' + 1$ is odd
3. up to changing the order of the generators of H, I_H is generated by the binomials

$$x_i^{\alpha_i} - x_{i+1} x_{i+e'+1}^{\alpha_{i+e'+1}-1} \qquad i = 1, 2, \ldots, e$$

   where we understand $x_j = x_{j-e}$ and $\alpha_j = \alpha_{j-e}$ if $j > e$ and α_i is the least integer such that $\alpha_i n_i$ has another factorization. Moreover, this is the unique binomial set of generators of I_H

4. The Row Factorization matrix $\mathrm{RF}(f)$ ($f = \frac{\mathrm{F}(H)}{2}$) has a special type which we call "Cascade Matrix", which we explain in the next slide.

(For $f \in \mathrm{PF}(H)$, $\mathrm{RF}(f) = (m_{ij})$ (Row Factorization Matrix) was introduced by A. Moscariello and defined so that $m_{ii} = -1$ for every i and $f = \sum_{j=1}^{e} m_{ij} n_j$. $\mathrm{RF}(f)$ is not unique in general)

I think this matrix $\mathbb{M}$ is the key of this talk.

$\mathbf{RF}(f) = \mathbb{M} = (m_{ij})_{1\le i,j\le e}$, where

$$m_{ij} = \begin{cases} -1 & i = j \\ 0 & i < j \le i + e' \\ \alpha_j - 1 & \text{otherwise.} \end{cases}$$

When $e = 9$,

$$\mathbb{M} = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & a_6 & a_7 & a_8 & a_9 \\ a_1 & -1 & 0 & 0 & 0 & 0 & a_7 & a_8 & a_9 \\ a_1 & a_2 & -1 & 0 & 0 & 0 & 0 & a_8 & a_9 \\ a_1 & a_2 & a_3 & -1 & 0 & 0 & 0 & 0 & a_9 \\ a_1 & a_2 & a_3 & a_4 & -1 & 0 & 0 & 0 & 0 \\ 0 & a_2 & a_3 & a_4 & a_5 & -1 & 0 & 0 & 0 \\ 0 & 0 & a_3 & a_4 & a_5 & a_6 & -1 & 0 & 0 \\ 0 & 0 & 0 & a_4 & a_5 & a_6 & a_7 & -1 & 0 \\ 0 & 0 & 0 & 0 & a_5 & a_6 & a_7 & a_8 & -1 \end{pmatrix}$$

Note that (1st row) – (second row) of $\mathbb{M}$ gives
$(a_1 + 1, -1, 0, 0, 0, a_6, 0, 0, 0)$, which gives
$X_1^{a_1+1} - X_2X_6^{a_6}$.

The following example is the beginning of this work from my viewpoint.
Example (Numata) (1) Let $H = \langle 11, 17, 19, 16, 15\rangle$. Then H is pseudo symmetric with $\mathrm{F}(H) = 40$ and

$$\mathrm{RF}(20) = \begin{pmatrix} -1 & 0 & 0 & 1 & 1 \\ 2 & -1 & 0 & 0 & 1 \\ 2 & 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & -1 \end{pmatrix}.$$

This example is the "smallest" (with smallest $\mathrm{F}(H)$) among PS and ACI with $e = 5$.
(2) For given e, odd, we define $\{m_1, m_2, \ldots, m_e\}$ by
$m_1 = 2^{e-1}, m_2 = m_1 + 1, m_3 = m_2 - 2, m_4 = m_3 + 4, m_5 = m_4 - 8$, and continue to $m_e = m_{e-2} - 2^{e-2}$. Then by suitable permutation of $m_1, \ldots, m_e$, $H = \langle m_1, m_2, \ldots, m_e\rangle$ is PS and ACI and also the smallest $\mathrm{F}(H)$ for e.
(3) We can actually change 2 to any positive integer a in (2). Then after suitable permutation, we get $\mathbb{M}$ in the previous slide with $a_1 = a$ and $a_2 = \ldots = a_e = a - 1$.

I list some key facts to prove our Main Theorem 1.
We call $F_i = X_i^{\alpha_i} - \prod_{j \neq i} X_j^{b_{ij}}$ a *critical binomial* and if I_H is ACI, then we must show that I_H is generated by critical binomials.

Fact 1. If for some $i \neq j$, if $X_i^{\alpha_i} - X_j^{\alpha_J}$ is a minimal generator of I_H, then H is not almost symmetric.
This fact follows from the previous works of K. Eto.

Given an **RF** matrix $\mathbb{M}$, the difference of i-th and j-th rows gives an element of I_H, which is called a **RF-relation**. It is shown by Herzog-Watanabe [HW] that if H is pseudo-symmetric, generated by 4 elements, then I_H is generated by RF-relations of $\mathbf{RF(F(}H\mathbf{)/2)}$.

Fact 2. For every H and for every minimal generator of I_H is an RF-relation of $\mathbf{RF(}f\mathbf{)}$ for some $f \in \mathbf{PF(}H\mathbf{)}$.
This fact plays an important role in our Proof.
But if we assume that H is pseudo-symmetric (which is not ACI), every minimal generator of I_H need not be generated by RF-relation of $\mathbf{RF(F(}H\mathbf{)/2)}$ (Matsuoka).

2nd Main Theorem; The ”Converse” to Main Theorem 1

Main Theorem 2. Let $\mathbb{M} := (m_{ij})$ be the “Cascade” matrix given as $\mathbf{RF}(f)$ in Main Thaorem 1. Let

$$(n_1, \ldots, n_e)^T = \mathrm{Det}(\mathbb{M}) \cdot \mathbb{M}^{-1} \mathbf{1}_e,$$

where we denote by $\mathbf{1}_e$ the column vector of length e, whose every entry is 1. Then we have

1. $\mathrm{Det}(\mathbb{M}) > 0$
2. if $\gcd(n_1, n_2, \ldots, n_e) = 1$, then $H = \langle n_1, n_2, \ldots, n_e \rangle$ is pseudo-symmetric and $k[H]$ is almost complete intersection as above with $\alpha_j = a_j + 1$ for $i = 1, 2, \ldots. e$.
3. $\mathrm{F}(H) = 2\,\mathrm{Det}(\mathbb{M})$

In the next slide, we will explain how we get $\mathrm{Det}(\mathbb{M}) > 0$.

Computation of $\mathbf{Det}(\mathbb{M})$

We can compute $\mathbf{Det}(\mathbb{M})$ for $\mathbb{M}$ in Main Theorem 2 by a combinatorial method.
We define $\mathbb{W}$ to be the $e \times e$ matrix permuting a_i of the Cascade matrix $\mathbb{M}$ in Main Theorrem 2 by **variable** w_i, so that
$P_e := \mathrm{Det}(\mathbb{W}) \in \mathbb{Z}[w_1, w_2, \ldots, w_e]$
and $\mathbf{Det}(\mathbb{M})$ is obtained by substituting w_i with a_i.

Theorem. Let $e = 2e' + 1 \geq 3$ be an odd integer. Consider the cycle with vertices $\{1, \ldots, e\}$ and edges $\{(i, i + e') \;:\; i = 1, \ldots, e\}$. Let $\mathcal{S} \subseteq \{1, \ldots, e\}$, and let $\mathcal{C} = \{1, \ldots, e\} \setminus \mathcal{S}$. The subgraph of the cycle induced by the vertex set $\mathcal{C}$ is a union of disjoint paths (if $\mathcal{C}$ is the whole cycle, we consider it a single path) of length $i_1, \ldots, i_r$. Then the monomial $x^{\mathcal{S}}$ appears in P_e with coefficient $e' - \sum_{j=1}^{r} \left\lceil \frac{i_j}{2} \right\rceil \geq 0$

Thank you very much.

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